

DELSBO, Sweden

WMO: 023470

Lat: 61.8264N

Lon: 16.5385E

Elev: 236

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.5	-1.6	-11.5	3.0	-5.9	-6.3	4.0	-1.6	18.0	35.0	16.0	34.4	1.9	270	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	79.2	65.3	75.6	63.2	72.4	61.3	67.1	75.6	65.0	72.6	63.0	69.8	6.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	89.7	70.2	62.0	83.8	68.3	60.0	78.0	66.4	31.7	75.6	30.0	72.8	28.6	69.9	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.9	12.7	11.1	-13.1	83.3	5.5	3.7	-17.1	86.0	-20.3	88.2	-23.4	90.2	-27.4	93.0		
(5)			-13.7	69.5	4.9	2.8	-17.3	71.5	-20.2	73.2	-22.9	74.8	-26.5	76.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	21.3	22.5	29.6	38.8	48.6	56.2	61.4	58.9	50.8	40.4	32.0	25.3		
	DBStd	15.81	10.74	10.82	8.68	5.82	6.15	5.30	4.75	4.99	5.47	6.48	7.36	10.48		
	HDD50	4404	889	771	631	337	101	9	0	2	54	304	541	765		
	HDD65	8942	1354	1191	1096	785	508	266	134	198	425	764	991	1230		
	CDD50	972	0	0	0	2	58	196	354	278	79	5	0	0		
	CDD65	35	0	0	0	0	0	3	23	9	0	0	0	0		
	CDH74	466	0	0	0	0	20	89	261	96	0	0	0	0		
	CDH80	62	0	0	0	0	2	10	44	6	0	0	0	0		
	WSAvg	4.4	4.2	4.1	4.9	4.8	5.0	4.7	4.2	3.7	3.9	4.4	4.3	4.3		
	PrecAvg	23.7	1.4	1.1	1.2	1.3	1.7	2.2	3.1	3.1	2.0	2.4	2.1	1.8		
	PrecMax	32.8	3.0	1.9	2.8	3.6	3.5	4.6	7.0	5.6	5.4	5.6	6.1	3.0		
(15) Precipitation	PrecMin	16.7	0.0	0.2	0.2	0.1	0.5	0.4	0.7	1.0	0.3	0.5	0.5	0.2		
	PrecStd	3.6	0.9	0.5	0.6	1.0	0.9	1.1	1.5	1.3	1.4	1.2	1.3	0.8		
	0.4%	DB	43.5	46.0	54.1	67.6	76.5	81.1	84.8	80.7	71.7	60.8	53.1	47.8		
	2%	MCWB	39.5	39.3	44.5	52.9	58.3	62.2	67.1	66.7	60.1	54.8	49.3	43.7		
		DB	40.9	42.8	49.1	60.1	70.3	76.3	80.6	76.8	67.4	56.6	48.7	43.8		
(23) Mean Coincident Wet Bulb Temperatures	5%	MCWB	37.6	38.0	41.6	48.8	56.5	61.7	66.3	65.1	58.5	51.3	45.5	40.2		
		DB	38.2	40.1	45.5	55.2	66.7	72.3	76.7	73.1	63.9	53.6	44.7	41.3		
	10%	MCWB	35.4	36.2	39.0	45.5	54.3	59.4	64.9	63.0	56.8	49.4	42.2	38.2		
		DB	35.3	36.9	42.4	51.4	62.5	68.7	73.1	69.6	61.0	50.9	42.1	38.5		
	10%	MCWB	33.0	33.8	37.1	42.9	52.0	57.7	62.5	61.3	54.7	47.6	40.4	36.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.1	40.4	45.0	54.1	61.3	65.5	70.5	69.6	61.9	55.5	49.8	44.2		
		MCDB	42.9	44.6	52.9	64.5	72.6	76.4	79.4	76.9	68.4	59.8	53.4	47.5		
	2%	WB	37.8	38.4	42.0	49.6	58.1	62.7	67.9	66.6	59.5	52.3	45.6	40.8		
		MCDB	40.4	42.2	48.3	59.2	67.9	72.9	76.8	73.6	65.7	55.2	48.2	43.3		
	5%	WB	35.9	36.4	39.5	46.4	55.3	60.7	65.9	64.4	57.7	50.2	42.9	38.2		
(33) Mean Daily Temperature Range	5% DB	MCDB	38.0	39.9	44.7	54.0	64.8	70.3	74.7	70.7	62.6	53.1	44.4	40.7		
		WB	33.2	34.0	37.2	43.7	52.9	58.6	63.9	62.5	55.8	47.9	40.7	36.0		
	5% WB	MCDB	34.8	36.6	42.0	50.3	61.4	66.8	71.0	68.0	59.8	50.5	42.0	38.1		
		WB	12.6	13.6	16.4	19.0	20.0	19.3	19.1	18.3	16.7	13.3	9.7	11.0		
	5% WB	MCDB	13.2	13.4	19.7	28.0	28.9	27.3	25.2	24.1	21.6	17.8	13.1	13.2		
(40) Clear-Sky Solar Irradiance	taub	MCWBR	11.5	10.0	13.4	17.3	16.3	14.5	13.2	13.5	13.6	12.9	11.0	11.3		
		MCDBR	13.1	12.8	18.9	25.5	26.5	25.0	22.7	20.4	19.7	16.4	12.3	13.0		
	Edh at Noon	MCWBR	11.5	9.9	13.2	16.4	15.6	14.0	12.6	12.3	13.1	12.5	10.5	11.4		
		MCDBR	13.1	12.8	18.9	25.5	26.5	25.0	22.7	20.4	19.7	16.4	12.3	13.0		
	RadAvg	71	272	750	1217	1550	1764	1641	1247	790	355	102	41			
(45) All-Sky Solar Radiation	RadStd	10	49	84	103	162	116	170	109	99	61	22	4			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-189	N/A	N/A	N/A	N/A

Nomenclature: See separate page